

@test_ema_learning

Appears in: DL-06

tests/test_ema_learning.py:50-110

"...PrecisionParts GmbH moved from 0.97 to 0.9760 on time and to 0.8808 after a five-day-late delivery, both matching the formula exactly and confirmed by @test_ema_learning."

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tests/test_ema_learning.py:50-110

50 def ema_expected(old, performance, alpha=0.2):
51     return round(min(1.0, max(0.0, alpha * performance + (1 - alpha) * old)), 4)
52
53
54 def run():
55     print("\n" + "=" * 65)
56     print("Betsy EMA Learning Loop Test")
57     print(f"Server: {BASE}")
58     print("=" * 65)
59
60     # — Target: PrecisionParts GmbH (SUP-004, highest baseline score 0.97)
61     SUP = "SUP-004"
62     SKU = "SKU-001" # supplied by SUP-004
63
64     baseline = supplier_score(SUP)
65     if baseline is None:
66         print(f"ERROR: supplier {SUP} not found – is the server running?")
67         return
68
69     print(f"\nSupplier: PrecisionParts GmbH ({SUP})")
70     print(f"Baseline score: {baseline:.4f}")
71
72     results = [("Baseline", baseline, "-", "-")]
73
74     # — Test 1: on-time delivery —————
75     print("\n[1] On-time delivery (lateness = 0 days)")
76     today = datetime.now().isoformat()
77     po1 = create_test_po(SUP, SKU, today)
78     patch(f"/api/purchase-orders/{po1}/status", status="in_transit")
79     r1 = patch(f"/api/purchase-orders/{po1}/status", status="delivered",
80               actual_delivery=today)
81
82     score_after_ontime = supplier_score(SUP)
83     expected_1 = ema_expected(baseline, 1.0)
84     match1 = "PASS" if abs(score_after_ontime - expected_1) < 0.0001 else "FAIL"
85     print(f"    Score: {baseline:.4f} -> {score_after_ontime:.4f} "
86           f"(expected {expected_1:.4f}) {match1}")
87     print(f"    Formula: 0.2x1.0 + 0.8x{baseline:.4f} = {expected_1:.4f}")
88     results.append(("After on-time delivery", score_after_ontime, expected_1, match1))
89
90     # — Test 2: 5-day late delivery —————
91     print("\n[2] Late delivery (+5 days)")
92     po2 = create_test_po(SUP, SKU, datetime.now().isoformat())
93
94     # Fetch the PO to get the server-assigned expected_delivery, then arrive 5 days late
95     po2_data = next(o for o in get("/api/purchase-orders") if o["po_id"] == po2)
96     expected_dt = datetime.fromisoformat(po2_data["expected_delivery"][:19])
97     actual_dt = (expected_dt + timedelta(days=5)).isoformat()
98
99     patch(f"/api/purchase-orders/{po2}/status", status="in_transit")
100    patch(f"/api/purchase-orders/{po2}/status", status="delivered",
101          actual_delivery=actual_dt)
102
103    score_after_late = supplier_score(SUP)
104    perf_late = max(0.0, 1.0 - 5 * 0.1) # = 0.5
105    expected_2 = ema_expected(score_after_ontime, perf_late)
106    match2 = "PASS" if abs(score_after_late - expected_2) < 0.0001 else "FAIL"
107    print(f"    Score: {score_after_ontime:.4f} -> {score_after_late:.4f} "
108          f"(expected {expected_2:.4f}) {match2}")
109    print(f"    Formula: 0.2x{perf_late:.1f} + 0.8x{score_after_ontime:.4f} = {expected_2:.4f}")
110    results.append(("After 5-day late delivery", score_after_late, expected_2, match2))
```

What it shows: The script that drives one on-time and one 5-day-late delivery against SUP-004 and asserts the score matches 0.9760 then 0.8808.

Source: tests/test_ema_learning.py:50-110 · image rendered by build_snippets.py